

- 5 The radionuclide $^{32}_{15}\text{P}$ decays by the emission of a single β -particle to a stable daughter nuclide. The half-life of $^{32}_{15}\text{P}$ radioisotope is 14.6 days. The activity of a $^{32}_{15}\text{P}$ source can be measured by recording the number of negative β -particles emitted by the source in five consecutive 10 s intervals as shown in Fig. 5.1.

interval	1	2	3	4	5
duration / s	10	10	10	10	10
number of β -particles	880	885	879	878	886

Fig. 5.1

- (a) Explain, with the help of a diagram, how the *random nature* of the radioactive decay and the *background radiation* will show up in the graph of variation with time of the activity.

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- (b) Show that the number of $^{32}_{15}\text{P}$ atoms in the source during the first 50 s is approximately 1.60×10^8 .

- (c) Another source contains a mixture of isotopes $^{32}_{15}\text{P}$ and $^{33}_{15}\text{P}$. The half-life of $^{33}_{15}\text{P}$ is 25.8 days. At time $t = 0$, 80% of the decays from this source is from $^{32}_{15}\text{P}$.

Show that the time taken for $^{33}_{15}\text{P}$ to contribute 70% of the decays is about 108 days.